JAIN Knowledge Campus, Jayanagar 9th Block, Bengaluru

DEPARTMENT OF COMPUTER SCIENCE AND IT

MCA PROGRAMME

ACADEMIC YEAR: 2026-2027

EXP: 8 - VISUALIZE REAL-TIME DATA FROM AZURE SYNAPSE ANALYTICS IN POWER BI

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SEM: III SPEC: SCT

SUBJECT: CLOUD ANALYTICS-LAB

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1 Aim

To visualize data from Azure Synapse Analytics in Power BI, using the student performance dataset already used in Experiments 04, 06, and 07, so that the dashboard can be checked against known values instead of against invented data. The experiment tests whether a Synapse serverless query can feed a Power BI report directly, without a dedicated SQL pool.

2 Objective and Theory

The objectives are to create an Azure Synapse workspace, query a comma-separated values file in Azure Data Lake Storage Gen2 through the Synapse serverless SQL pool, connect Power BI to that serverless endpoint, build dashboard visuals from the query result, and record the connection mode actually used.

Azure Synapse Analytics is a cloud analytics service that combines data warehousing and big data analytics. Its serverless SQL pool, called Built-in in Synapse Studio, lets a user query files in a data lake with Transact-SQL (T-SQL), without provisioning a dedicated pool [1]. The OPENROWSET function reads a file at a given storage URL and returns its rows as a table, with column types declared in a WITH clause [2]. Power BI is a business analytics service for creating interactive reports and dashboards. DirectQuery is a Power BI connection mode in which visuals query the source at load time, instead of importing a copy of the data [3]. In the Power BI Service Get Data flow used for this experiment, the semantic model actually created was Import, as verified in the model view, not DirectQuery. With Import mode, updated data reaches the report through a semantic model refresh, and that refresh step is the real-time insight mechanism demonstrated here, as in Experiment 07.

The dataset contains 50 student records with scores in mathematics, reading, and writing, attendance rate, study hours per week, and a final grade from A to F. Two fields have missing values in the source file, attendance rate and study hours per week, each with 45 non-null values out of 50. Those missing values are kept as null values, and averages are calculated over the non-null values, which is stated here so that the averages in Section 5 can be reproduced exactly.

3 Architecture Flow

Data flows from Azure Data Lake Storage Gen2 through a Synapse serverless SQL query into Power BI, as shown in Figure 1.1. Table 1.1 lists the resources used. No dedicated SQL pool and no Spark pool were created for this experiment.

Table 1.1 Resources used in Experiment 08

Component	Resource name	Role
Resource Group	25MCAR0141	Logical container for all experiment resources.
Storage Account	calablake141	Azure Data Lake Storage Gen2, rawdata container, student_performance.csv.
Synapse Workspace	calab-synapse-141	Synapse workspace, Central India, provisioning Succeeded.
Synapse Serverless	calab-synapse-141-ondemand.sql.azuresynapse.net	Built-in serverless SQL pool, database master, OPENROWSET query.
Power BI	Power BI Service, Import	Report Exp08_PowerBI_Synapse, five visuals, semantic model refresh.

Source: Azure resource list and Power BI report record, verified 2026-11-01.

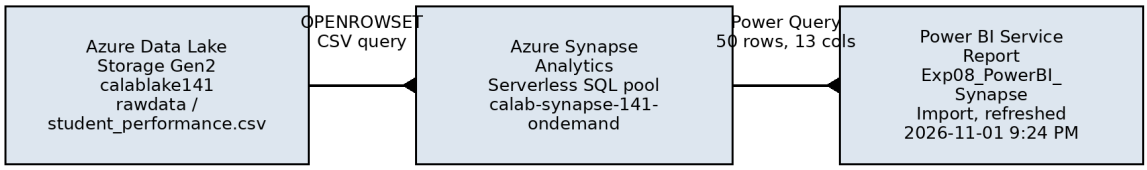


Figure 1.1 Experiment 08 data flow

Data Lake Storage to Synapse serverless SQL pool to Power BI, with Import refresh.

4 Procedure

Step 1 (Synapse workspace creation). Create Synapse workspace calab-synapse-141 in resource group 25MCAR0141, location Central India, with primary storage account calablake141 and file system rawdata. The workspace provisioning state returned Succeeded, with serverless endpoint calab-synapse-141-ondemand.sql.azuresynapse.net and dedicated endpoint calab-synapse-141.sql.azuresynapse.net. No dedicated SQL pool was created, so the dedicated endpoint has no pool behind it, and no Spark pool was started.

Step 2 (Synapse Studio attempt and access fix). In Synapse Studio, signed in as abhishek.25005296@jainuniversity.ac.in, open a new SQL script with Connect to Built-in. Synapse Studio showed Unable to connect to serverless SQL pool because of a network or firewall issue, Run remained disabled, and Use database showed No database available. The workspace firewall was then fixed in stages: a single IP rule, a

range rule for 87.81.0.0 to 87.81.255.255, and a temporary allow-all rule AllowAllTemp, and the user already held the Synapse Administrator role, confirmed by a RoleAssignmentAlreadyExists result. A retry still showed the same serverless connection failure, and the 403 dialog for workspace resources no longer appeared, but Run stayed disabled. The view vw_StudentPerformance was therefore not created in Synapse Studio, and that failure is reported here instead of being hidden. A Storage Blob Data Reader role was also granted to the user on calablake141 on 2026-11-01, so that a serverless query could read the CSV file.

Step 3 (Power BI direct query). In Power BI Service, use Get data, Azure Synapse Analytics SQL, with server calab-synapse-141-ondemand.sql.azuresynapse.net, database master, and Organizational account authentication, signed in as abhishek.25005296@jainuniversity.ac.in. Use the SQL statement in synapse_serverless_query.sql in the GitHub folder, which selects from OPENROWSET over https://calablake141.dfs.core.windows.net/rawdata/student_performance.csv, with parser version 2.0, header row true, and the 13 columns declared in the WITH clause. Power Query preview reported 50 rows and 13 columns, completed in 1.52 seconds, as shown in Screenshot 3.

Step 4 (Dashboard visuals). Build five visuals from table StudentData: KPI cards for total students and average scores; a clustered column chart of average mathematics, reading, and writing scores by FinalGrade; a bar chart of student count by FinalGrade; a donut chart of student share by Gender; and a table of FinalGrade, student count, and average scores. The queries for these visuals are in dashboard_queries.sql in the GitHub folder.

Step 5 (Save and verify). Save the report as Exp08_PowerBI_Synapse in My workspace. Verify every visual shows data, the grade counts sum to 50, the gender counts sum to 50, and the table totals match the KPI cards. The model view shows Storage mode Import and Data refreshed 11/1/2026, 9:24:21 PM, and the workspace list shows the report and semantic model refreshed 11/1/2026, 9:24:22 PM. The connection mode actually used was therefore Import, not DirectQuery, and the report record in Section 5 states that plainly.

5 Proof of Work

The GitHub repository for this experiment is https://github.com/25MCAR0141/Cloud-Analytics-Azure/tree/main/Lab/Exp_08 . The repository contains the README, architecture diagram, Power BI screenshots, Synapse query, and dashboard queries, without connection secrets. Table 1.2 records the Power BI execution, and Table 1.3 records the dashboard data calculated from the 50 rows returned by the Synapse serverless query.

Table 1.2 Power BI execution, Experiment 08

Item	Value
Report	Exp08_PowerBI_Synapse, id c179d7f7-ff78-424f-9008-9a7f501f0a63
Semantic model	Exp08_PowerBI_Synapse, id 58d8ce9b-ac91-45e9-9dab-dfa9f835608b
Workspace	My workspace, id cfa9da9e-3370-4b81-b46c-b14de945b284
Source	Synapse serverless calab-synapse-141-ondemand.sql.azuresynapse.net, database master
Query	OPENROWSET over calablake141 rawdata student_performance.csv, 50 rows, 13 columns
Connection mode	Import, verified in model view, refreshed 11/1/2026, 9:24:21 PM
Authentication	Organizational account, abhishek.25005296@jainuniversity.ac.in

Source: Power BI Service report and model view, 2026-11-01.

Table 1.3 Dashboard data, average scores by FinalGrade

FinalGrade	Students	Avg MathScore	Avg ReadingScore	Avg WritingScore
A	16	87.50	90.38	88.63
B	12	73.17	76.58	74.83
C	10	57.60	53.80	53.10
D	6	46.67	43.67	42.17
F	6	38.67	34.00	36.33

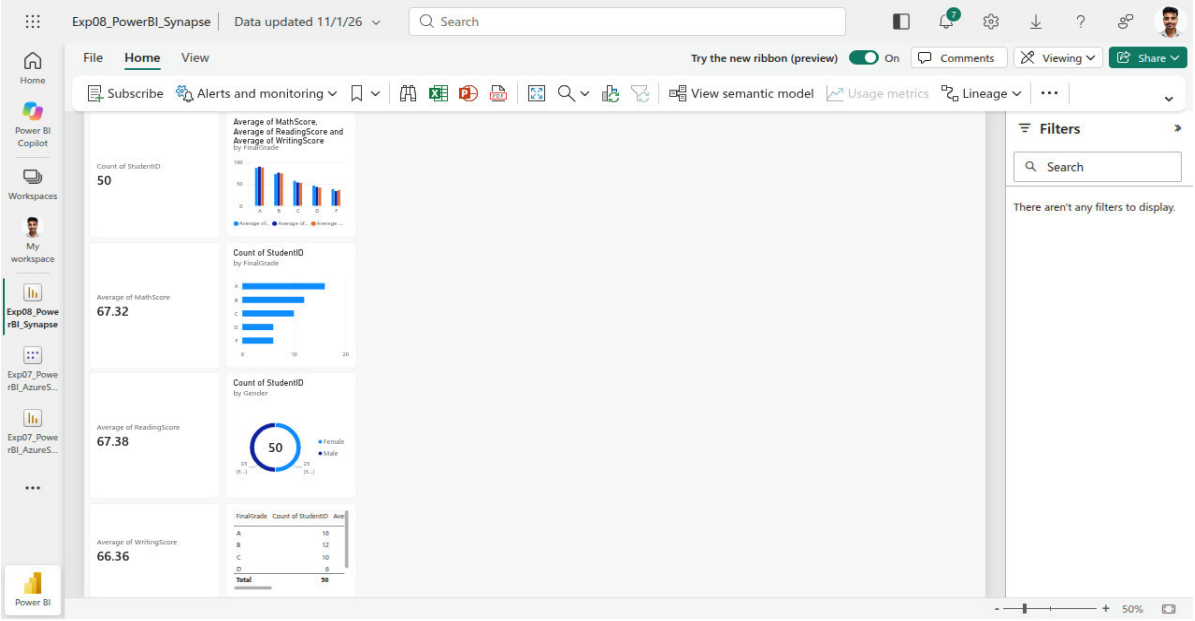
Source: Synapse serverless query result, 50 rows, verified 2026-11-01. Overall averages are MathScore 67.32, ReadingScore 67.38, and WritingScore 66.36. Underlying sums are Math 3366, Reading 3369, and Writing 3318.

Power BI report record. The saved Power BI Service report is Exp08_PowerBI_Synapse, at https://app.powerbi.com/groups/cfa9da9e-3370-4b81-b46c-b14de945b284/reports/c179d7f7-ff78-424f-9008-9a7f501f0a63/04bf88715006c59c07d8?experience=power-bi , with semantic model Exp08_PowerBI_Synapse, refreshed

2026-11-01 at 9:24 PM. Page 1 contains five verified visuals: KPI cards for student count 50 and average scores 67.32, 67.38, and 66.36; a clustered column chart of average scores by FinalGrade; a clustered bar chart of student count by FinalGrade; a donut chart of student share by Gender, Female 25 and Male 25; and a table of FinalGrade, student count, and average scores, with total 50, 67.32, 67.38, and 66.36.

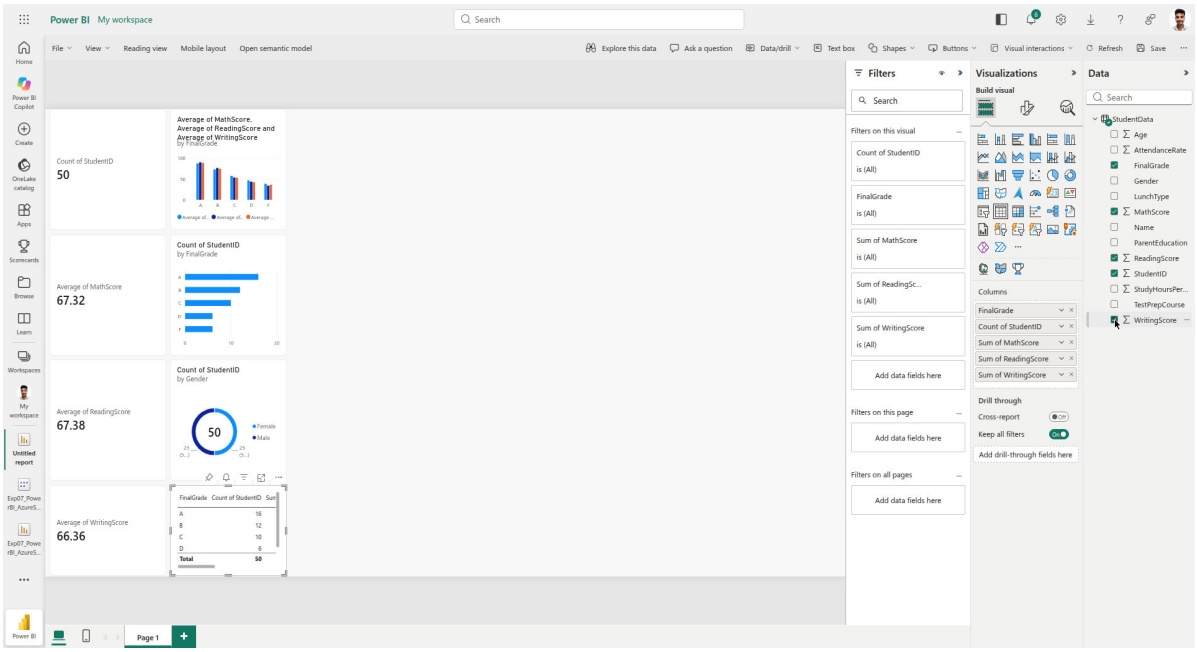
6 Screenshots

The screenshots in this section are real browser screenshots of the saved Power BI Service report, captured from the signed-in Power BI session on 2026-11-01, not replotted charts and not cropped chart images. The report title Exp08_PowerBI_Synapse is visible in each report screenshot, and the report URL is given in Section 5. The browser capture shows the Power BI viewport, not the operating system browser chrome, so the address bar and tabs are not drawn inside the images, as in Experiment 07. Figure 1.1 in Section 3 remains the architecture diagram, and Tables 1.2 and 1.3 in Section 5 remain the execution and data records; this section is the visual proof of the dashboard itself.



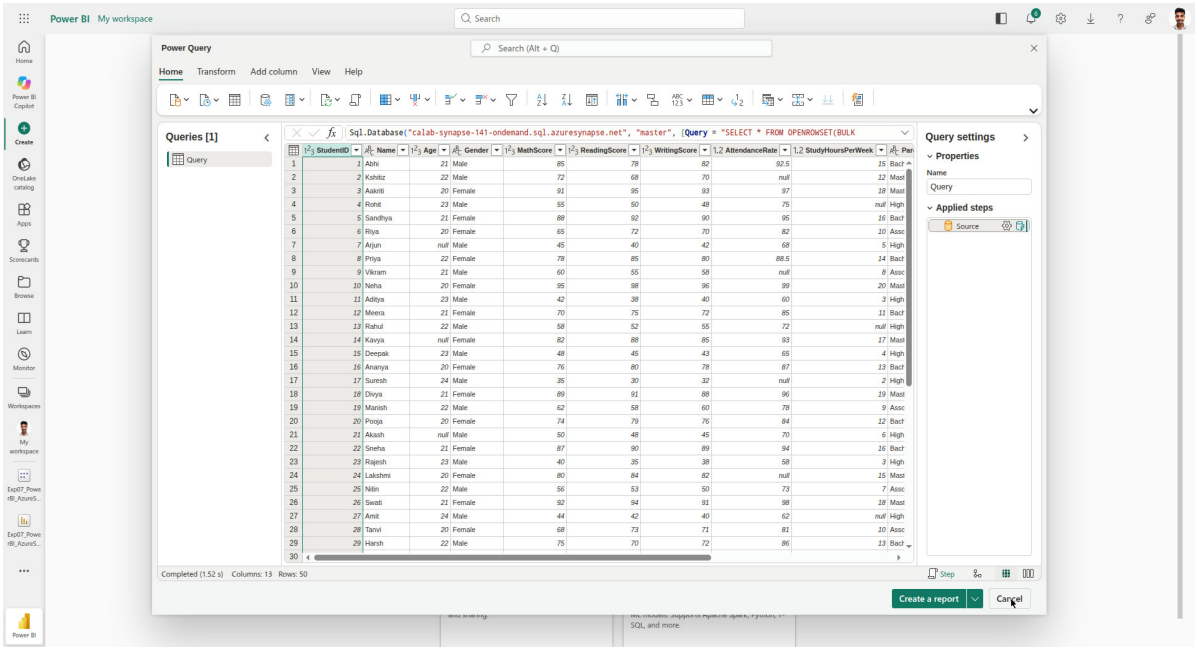
Screenshot 1 Power BI Service report Exp08_PowerBI_Synapse, Page 1, reading view at 50 percent

All five visual groups in one view: KPI cards for student count 50 and average scores 67.32, 67.38, and 66.36, clustered column chart by FinalGrade, bar chart by FinalGrade, donut chart by Gender, and FinalGrade table with total 50.



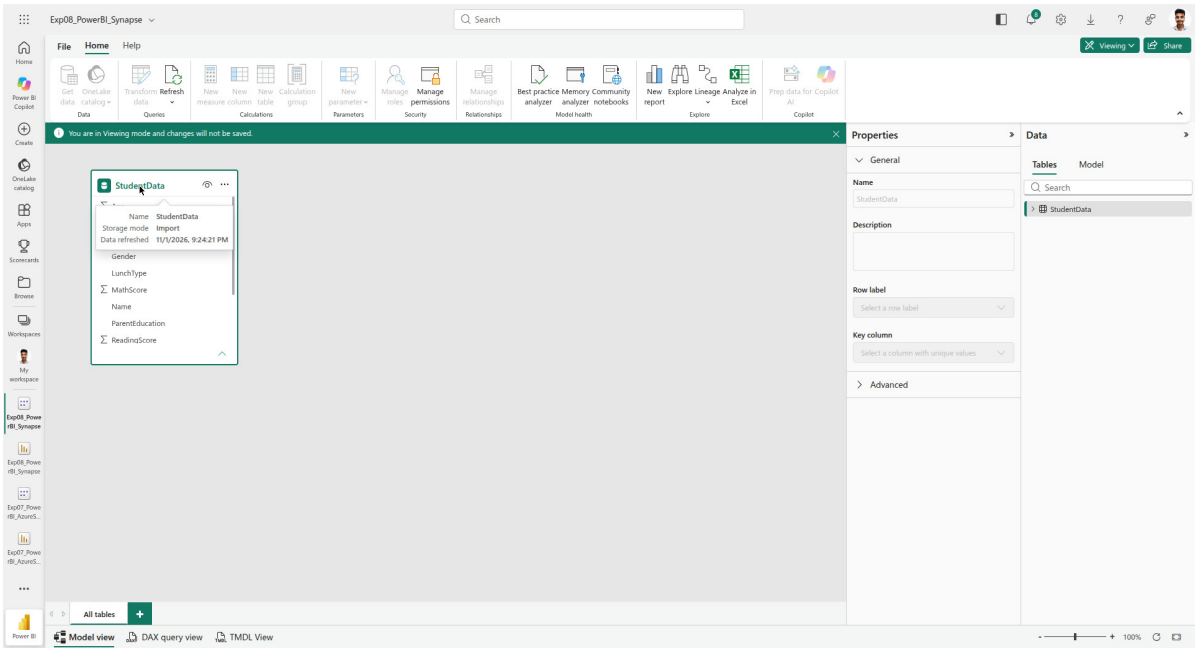
Screenshot 2 Power BI Service report Exp08_PowerBI_Synapse, Page 1, editing view

Same five visuals at larger size, with KPI cards, column chart, bar chart, donut chart showing 50, and the FinalGrade table, in the report editor before saving.



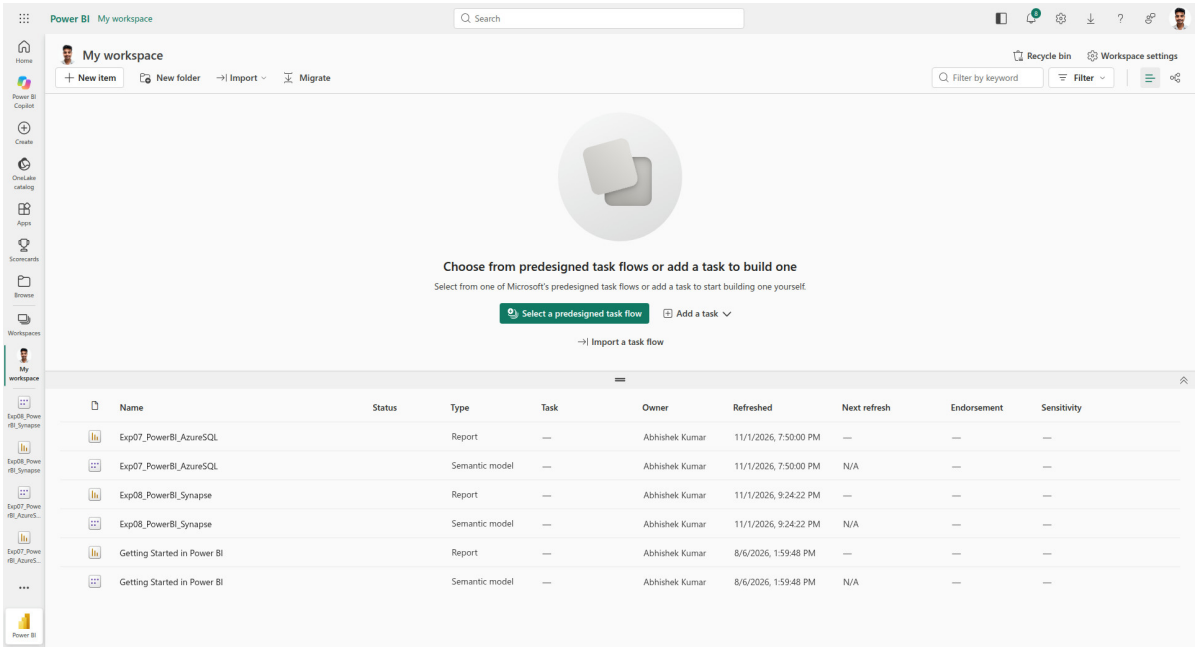
Screenshot 3 Power Query preview for the Synapse serverless query

Query Sql.Database for calab-synapse-141-ondemand.sql.azuresynapse.net, database master, with the OPENROWSET statement, preview completed in 1.52 seconds, Columns 13, Rows 50.



Screenshot 4 Power BI semantic model view for Exp08_PowerBI_Synapse

Table StudentData, Storage mode Import, Data refreshed 11/1/2026, 9:24:21 PM, which is the Import mode evidence.



Screenshot 5 Power BI My workspace list for Exp08_PowerBI_Synapse

Report and semantic model Exp08_PowerBI_Synapse, Owner Abhishek Kumar, Refreshed 11/1/2026, 9:24:22 PM, with the Experiment 07 report listed above for comparison.

Cost note. No dedicated SQL pool and no Spark pool were created for this experiment, so no hourly pool charge applies. The Synapse workspace itself and the serverless queries over a 50-row file are minimal, and the temporary firewall rule AllowAllTemp should be removed after submission, to restore the previous firewall posture.

References

[1] Microsoft, “Azure Synapse Analytics documentation,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/synapse-analytics/

[2] Microsoft, “OPENROWSET (Transact-SQL),” Microsoft Learn. Available: https://learn.microsoft.com/en-us/sql/t-sql/functions/openrowset-transact-sql

[3] Microsoft, “Use DirectQuery in Power BI Desktop,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/power-bi/connect-data/desktop-directquery-about